MANISH CHHETRI

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PROFILE

Software engineer with experience building distributed, event-driven backend systems and AI-agent pipelines on GCP and AWS. Skilled in microservices, Kubernetes and scalable cloud architecture, with a research background in AI and NLP.

Interests: Artificial Intelligence, Cloud Computing, Machine Learning, NLP, Computer Vision, IoT.

EDUCATION

Western Region Campus, Institute of Engineering, TU, Nepal *2019 – 2024*
B.E. in Electronics, Communication and Information Engineering
Final Grade: First Division *Percentage: 72.55%*
Thesis Title:
Image Captioning in Nepali Language

Global Collegiate School, Nepal
Science, National Examination Board (NEB) *2017–2019*
Final Grade: First Division *CGPA: 3.45/4.0*

SKILLS AND COMPETENCIES

Languages:	Python, C/C++, JavaScript (Node.js), Arduino
Machine Learning:	Transformers, LLMs, CNNs, RNNs
Web Development:	Node.js, TypeScript, Express, NestJS, React.js, Next.js
Database:	PostgreSQL, MySQL, MongoDB
DevOps & Cloud:	Docker, Kubernetes, CI/CD (GitHub Actions), AWS (EC2, S3, Lambda, VPC), GCP(Cloud Run, GKE, Pub/Sub,Cloud Tasks, Cloud Functions)
Hardware:	Arduino, SIMmodule, GPS, Bluetooth, VHDL (Embedded Programming)
Strengths:	Communication, Team Leadership, Time Management, Project Coordination

PUBLICATIONS

Nepali Image Captioning: Generating Coherent Paragraph-Length Descriptions Using Transformer *Journal of Soft Computing Paradigm, 6(1), 70-84 2024*
CO-AUTHOR
- Subedi, N., Paudel, N., Chhetri, M., Acharya, S., & Lamichhane, N
- DOI: 10.36548/jscp.2024.1.006

Drowsiness and Crash Detection Mobile Application for Vehicle's Safety *Journal of IoT in Social, Mobile, Analytics, and Cloud, 6(1), 54-66 2024*
CO-AUTHOR
- Subedi, N., Paudel, N., Chhetri, M., Acharya, S., & Lamichhane, N.
- DOI: 10.36548/jismac.2024.1.005

RESEARCH EXPERIENCE

Nepali Image Captioning *2024*

- Created Nepali Paragraph Captioning Dataset using Beautiful Soup for web scraping and manual annotation.
- Developed a web app using Node.js and React, deployed models to Hugging Face for inferencing.
- Built image captioning models using ResNet50, ResNet152, InceptionV3 for feature extraction and LSTM/Transformer for Nepali caption generation.
- Authored a comprehensive scientific paper and presented at an international conference, receiving recognition for its impact.
- Using LSTM
- Using Transformer

Crash and Drowsiness Detection System *2023*

- Integrated Arduino, GPS, GSM, Accelerometer, and Bluetooth module to detect collisions and notify rescue centers.
- Implemented false notification prevention using a mobile app connected to the crash system.
- Implemented facial detection system to monitor driver drowsiness.
- Authored and submitted a manuscript, serving as corresponding author.
- CODE

CONFERENCE/WORKSHOP ATTENDED

Recent Trends on Science, Technology and Innovation 2024 (RTSTI) *Institute of Forestry, Pokhara Campus*
PRESENTER *2024*

- Presented a comprehensive research on Vehicle's Crash Detection and Alert System.
- Received positive feedback for innovative work on combining hardware with AI.

INDUSTRIAL EXPERIENCE

Maitri Services *Lalitpur, Kathmandu Nepal*
Software Engineer *Oct. 2025 – Present*

- Designed and built a distributed, event-driven backend on GCP (Cloud Run, GKE, Pub/Sub, Cloud Tasks, Cloud Functions), with services communicating through both asynchronous messaging and direct APIs.
- Designed orchestration agents and sub-agents that process large documents (100+ pages) across 33+ legal and financial document types, running asynchronously and in parallel to extract structured content and automate workflows between businesses and their clients.
- Built an AI-agent-driven integration platform that connects third-party finance services to the core system through a generic connector framework, so new integrations need minimal custom code.
- Integrated Recall.ai to manage meeting bots across Google Meet, Zoom and Microsoft Teams, and OpenFGA for fine-grained authorization.
- Built AI-triggered workflows for a B2B2C platform that turn meeting context and uploaded financial and legal documents into automated follow-up actions, plus a companion mode that assists both clients and businesses within the system.

- Designed scalable, auditable backend systems, improving performance through database optimization and CI/CD improvements.

Swivt Technology *Lalitpur, Kathmandu Nepal*
Full Stack Software Engineer *Nov. 2023 – Aug. 2025*

- Managed server configurations and resolved production issues by analyzing logs, reducing downtime by ~30%
- Delivered 5+ international client projects with high-quality results and consistently positive feedback.
- Added key features to existing systems, including Payment Systems, Admin Panel, CMS, CRM, WebRTC, and role-based access control.
- Containerized applications, optimized Nginx, managed Cloudflare DNS/domains, and deployed RAG-based transformer models for a medical insurance system in a cloud-native setup.

EnterLeaf *Kuleshwor, Kathmandu Nepal*
Microservice Backend Developer *Jul. 2023 - Nov. 2023*

- Designed and developed different services using NestJS, leveraging NATS for message brokering, enabling a robust event-driven architecture.
- Integrated an API Gateway to handle and route requests efficiently from both mobile and web browser clients, ensuring seamless communication across services
- Implemented a CI/CD pipeline using GitHub Actions, streamlining the deployment process and ensuring continuous integration and delivery for rapid and reliable updates.

UpStartPrep *Knoxville, Tennessee, United States*
Backend Developer *Dec. 2022 -Jun. 2023*

- Creating and improving application programming interfaces (APIs), architecture, and servers, and databases (SQL) for E-Learning Platforms.
- Implementation of Payment System (Stripe) and Used Cloud Services like AWS.
- Implementation of microservices-based architecture, incorporating containerization and CI/CD pipeline for efficient development.

AWARDS AND HONOURS

Robo Soccer Competition *National Innovation Center Nepal, Pokhara*
First Position *Jan. 2019*

- Designed and developed an autonomous robot in a team of four that could play soccer, kick the ball, and defend the goal.

Battle for Speed *VECTOR – Second National Technical Festival, WRC*
Second Position *Jan. 2019*

- Developed an automatic line-following robot using Arduino programming, focusing on speed and precision.

Datathon Competition *I-CES, IOE Western Region Campus*
First Position *Jan. 2019*

- Collaborated in a two-member team to preprocess data, train a machine learning model with 91% accuracy, and present the results successfully.